Code No: 135CE **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, January/February - 2023
ENVIRONMENTAL ENGINEERING
(Common to CE, CSE, IT)

Time: 3 Hours **Max.Marks:75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Write down fire demand along with equation for calculation.	[2]
b)	Write down any three drinking water standards.	[3]
c)	Define coagulation.	[2]
d)	How will calculate chlorine demand?	[3]
e)	What is combined flow?	[2]
f)	What are the uses of air valves?	[3]
g)	Write down the advantages one pipe systems of plumbing.	[2]
h)	Draw a manhole along with components.	[3]
i)	Write down the purpose of slugged drying.	[2]
j)	Can we use soak pits in the sandy soils area, justify your answer.	[3]

PART – B

(50 Marks)

2. The population of a locality as obtained from census report is as follows.

Census Year	1941	1951	1961	1971	1981	1991	2001	2011	2021
Population	8000	12000	17000	22500	29000	37500	47000	57000	66500

Estimate the population of the locality in the years 2021, 2031 and 2041, by
a) Incremental increase method.
b) Arithmetical increase method. [10]

OR

3. Discuss the following as they relate to water supply
a) Intermittent supply versus continuous supply
b) Metered supply versus non-metered supply
c) Single supply versus dual supply. [3+4+3]

4. Compare rapid and slow sand filters under the following heads:
a) Rate of filtration
b) Quantity of wash water
c) Method of cleaning
d) Effective size of sand
e) Depth of bed for gravel and sand. [10]

OR

5.a) Describe with a neat sketch the working of a rapid sand filter (gravity type). What are its advantages over the slow sand filter?

b) Design a set of rapid gravity filters for treating water required for a population of 50,000; the rate of supply being 180 litres per head per day. The filters are rated to work at 5200 litres per square metre per day. [5+5]

6. Calculate the storage required to supply the demand shown in the following table if the inflow of water to the reservoir is maintained at a uniform rate throughout 24 hours.[10]

Time (hours)	00-04	04-08	08-12	12-16	16-20	20-24
Demand (million litres)	0.48	0.87	1.33	1.00	0.82	0.54

OR

7. Write short notes on the following:
a) Sluice valve; b) Reflux valve; c) Fire hydrant; d) Air valve; e) Domestic water meter. [10]

8. Write short notes on:
a) B.O.D. and C.O.D.
b) Aerobic and anaerobic bacteria.
c) Method of determination of B.O.D. [3+3+4]

OR

9.a) What is Oxygen sag curve in stream pollution?

b) Explain self-purification of streams and indicate how sun-light helps in self-purification. [5+5]

10. Write short notes on:
a) Sludge conditioning
b) Sludge digestion
c) Sludge digestion tank. [3+3+4]

OR

11.a) Explain the theory of purification of sewage by trickling filters. Describe the constructional details with neat sketches.

b) It is proposed to install a two stage trickling filter for treatment of 3.8 m.l.d. of raw sewage. The B.O.D. = 200 ppm. The following particular are given:
i) The filter loading is 1100 gm/m^3 of B.O.D.
ii) Recirculation ratio is 1 for each filter.
iii) 35% B.O.D. removed in primary tank.
iv) Total volume is equally divided into two filters.
What will be the BOD of the effluent? [5+5]

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